

MAST32001 Computational Statistics - Exercise Sheet 4

Bootstrap for structured data and cross-validation

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This sheet continues the material on bootstrap methods for regression and dependent data and on cross-validation/model selection.

For every resampling problem, do not only compute a number. State **what is being resampled, what is kept fixed, and why the resampling scheme is appropriate for the question.**

Submission

Solutions must be submitted in **Moodle no later than 30.09.2026 at 12:00.**

Please submit your solutions in the **submission area corresponding to your exercise group.** Make sure that you submit to the correct exercise group.

Exercise points

Exercise points are awarded only in connection with the corresponding exercise session.

To receive the point(s) for a task, you must:

1. submit your solution by the deadline;
2. attend the exercise session in which the task is discussed; and
3. be willing to **present, explain, and discuss your solution** during the exercise session if asked.

Submitting a solution without attending the corresponding exercise session does **not** earn exercise points.

Your solution does not need to be perfect in order to be discussed. The purpose of the exercise sessions is also to compare approaches, identify difficulties, and learn from different solutions.

Use of generative AI

The use of **generative AI tools to solve the exercise tasks is not permitted.**

The purpose of the exercises is to develop your own ability to translate statistical ideas into algorithms, implement them in R, investigate numerical behaviour, and explain your reasoning.

The submitted solutions, including the reasoning and R code, must therefore reflect **your own work.**

Exercise 1 — Is there serial correlation?

The annual Nile river-flow series is available in R as `Nile`. We want to investigate whether consecutive observations show positive serial dependence.

1. Plot the series and its sample autocorrelation function.

2. Use the lag-one sample correlation

$$T = \text{cor}(X_1, \dots, X_{n-1}, X_2, \dots, X_n)$$

as a test statistic for

$$H_0 : \text{the observations are iid}$$

against positive serial dependence.

3. Consider the following possible ways to obtain a null distribution for T :
 - a. randomly permute the observed values;
 - b. ordinary iid bootstrap of the observed values;
 - c. moving-block bootstrap;
 - d. fit an AR(1) model to the original series and simulate from the fitted model.

For each method, decide whether it is appropriate for this particular null hypothesis. Explain your answer before doing any simulation.

4. Carry out a permutation test with $B = 4999$ random permutations. Use

$$p = \frac{1 + \sum_{b=1}^B I(T_b^* \geq T_{\text{obs}})}{B + 1}.$$

5. Interpret the result. What exactly does rejection tell us, and what does it **not** tell us?
6. Suppose instead that the goal were to estimate the standard error of the sample mean while allowing serial dependence. Would the same permutation scheme be appropriate? Which resampling idea from the lectures would be more natural?

Exercise 2 — Regression bootstrap under heteroscedasticity

Download the file `Week4Ex2.txt` from Moodle and consider the data it contains.

We are interested in the uncertainty of the slope coefficient.

1. Plot x against y and describe what you see. Focus on the conditional mean and the variability. Next, plot residuals against fitted values. What feature of the usual homoscedastic linear model looks questionable? We are in the following interested in the uncertainty of the slope coefficient.
2. Compare these three bootstrap schemes:
 - residual bootstrap;
 - pairs bootstrap;
 - wild bootstrap with

$$y_i^* = \hat{y}_i + \hat{e}_i v_i, \quad P(v_i = -1) = P(v_i = 1) = 1/2.$$

For each method, describe what data-generating mechanism it is pretending to reproduce.

3. Implement all three methods with $B = 3000$ and estimate the bootstrap standard error of the slope.
4. Repeat the entire experiment below for 300 independently generated datasets. For each dataset, compute a nominal 95% normal bootstrap interval for the slope using each method. Estimate the empirical coverage probability of the true slope 1.5.
5. Which bootstrap scheme is most defensible for this data-generating process? Explain the result in terms of what structure is preserved or destroyed.

Exercise 3 — Cross-validation, model complexity, and selection bias

We use the `swiss` data. The response is `Fertility`; the other variables are possible predictors.

1. Inspect the data using suitable plots and summaries. Do not fit all candidate models immediately. Which predictors appear potentially useful?
2. Consider the following nested models:

$$M_1 : \text{Fertility} \sim \text{Education},$$

$$M_2 : \text{Fertility} \sim \text{Education} + \text{Agriculture},$$

$$M_3 : \text{Fertility} \sim \text{Education} + \text{Agriculture} + \text{Examination},$$

$$M_4 : \text{Fertility} \sim \text{Education} + \text{Agriculture} + \text{Examination} + \text{Catholic},$$

$$M_5 : \text{Fertility} \sim \text{Education} + \text{Agriculture} + \text{Examination} + \text{Catholic} + \text{Infant.Mortality}.$$

Before computing anything, describe qualitatively how you expect training error, bias, and variance to change from M_1 to M_5 .

3. Perform 10-fold cross-validation using the **same folds for all five models**. Compute the CV MSE for each model.
4. Repeat the complete 10-fold CV 100 times with new random fold assignments. For each repetition, record which model has the smallest CV MSE. How stable is the selected model?
5. Fit all five models to the full dataset and compare AIC and BIC. Do AIC, BIC, and repeated CV necessarily choose the same model? Why not?
6. A colleague reports the smallest CV MSE from step 3 as an unbiased estimate of the prediction error of the whole model-selection procedure. Explain the problem with this statement.
7. Describe how nested cross-validation could be used to estimate the predictive performance of the complete procedure “choose one of $M_1, \dots, M_5$ by CV and then fit it.”

Exercise 4 — A theoretical CV problem: what are you estimating?

Suppose

$$(X_i, Y_i), \quad i = 1, \dots, n,$$

are iid observations and a learning algorithm A maps a training sample $\mathcal{D}$ to a predictor $\hat{f}_{\mathcal{D}}$.

Let

$$R_m = E_{\mathcal{D}_m} E_{(X_0, Y_0)} \left[L\{Y_0, \hat{f}_{\mathcal{D}_m}(X_0)\} \right]$$

denote the expected prediction risk of the algorithm when trained on m observations.

1. In K -fold CV with equally sized folds, how many observations are used to fit each model? Which risk R_m is CV most directly connected to?
2. Explain why this creates a possible bias when we ultimately refit the selected algorithm using all n observations.
3. Compare 5-fold CV and LOOCV in terms of:
 - training-set size;
 - number of fitted models;
 - similarity of the fitted training samples;
 - likely computational cost;
 - why the variance of the CV estimate is not determined only by the number of validation observations.
4. A student proposes the following workflow:
 - standardize every predictor using the mean and standard deviation from the complete dataset;

- run 10-fold CV;
- choose the best tuning parameter.

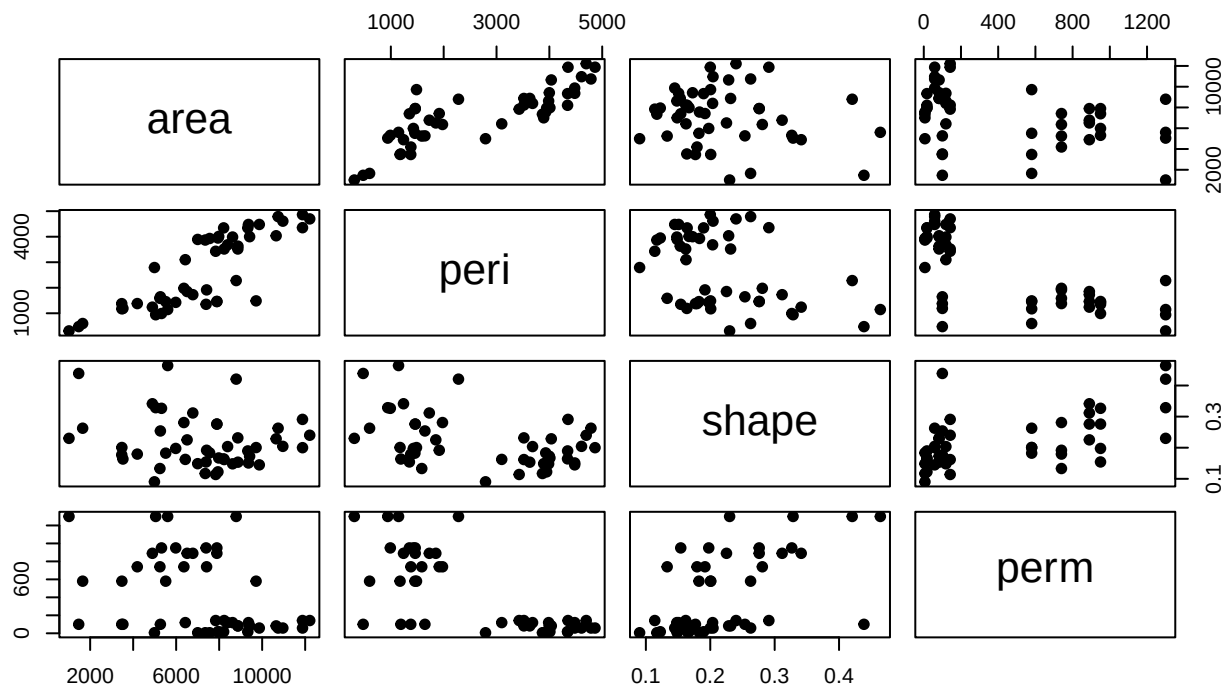
Is the CV estimate cleanly out-of-sample? Explain and give the correct workflow.

- Suppose we have repeated measurements from 50 patients, with 8 measurements per patient. The scientific goal is prediction for **new patients**. Should we assign the 400 observations independently to random CV folds? Explain.
- In one or two sentences each, state the main conceptual difference between:
 - ordinary K -fold CV;
 - nested CV;
 - AIC;
 - BIC.

Exercise 5 — Prediction after model selection

We use the `rock` data in R. The response `perm` measures permeability, and `area`, `peri`, and `shape` are possible predictors.

```
data(rock)
pairs(rock, pch = 19)
```



```
summary(rock)
```

##	area	peri	shape	perm
## Min.	: 1016	Min. : 308.6	Min. : 0.09033	Min. : 6.30
## 1st Qu.	: 5305	1st Qu.: 1414.9	1st Qu.: 0.16226	1st Qu.: 76.45
## Median	: 7487	Median : 2536.2	Median : 0.19886	Median : 130.50
## Mean	: 7188	Mean : 2682.2	Mean : 0.21811	Mean : 415.45
## 3rd Qu.	: 8870	3rd Qu.: 3989.5	3rd Qu.: 0.26267	3rd Qu.: 777.50
## Max.	: 12212	Max. : 4864.2	Max. : 0.46413	Max. : 1300.00

Consider the candidate models

$$M_1 : \text{perm} \sim \text{area},$$

$$M_2 : \text{perm} \sim \text{peri},$$

$$M_3 : \text{perm} \sim \text{shape},$$

$$M_4 : \text{perm} \sim \text{area} + \text{peri},$$

$$M_5 : \text{perm} \sim \text{area} + \text{peri} + \text{shape}.$$

1. Randomly split the data into 80% training observations and 20% test observations. Set the test data aside and do not use them for model selection.
2. Using only the training data, choose one of $M_1, \dots, M_5$ by 5-fold cross-validation.
3. Refit the selected model on the complete training set and compute its test MSE on the untouched test set.
4. Now make the following deliberately bad analysis: choose the model with the smallest **test-set** MSE and report that minimum test MSE as the estimated prediction error. Explain why this estimate is optimistically biased.
5. Repeat steps 1–3 for 200 random train/test splits. Record
 - which model is selected by CV;
 - its test MSE.

Summarize the selection frequencies and the distribution of test MSEs.

6. A colleague says: “The model selected most often must be the true model.” Explain why this conclusion does not follow from the experiment.
7. Explain the distinction between the variability seen here and the bias–variance trade-off of a prediction method.